VRMTS-RAG System Evaluation Report

Retrieval-Augmented Generation for Anatomy Tutoring

Model: Llama 3.2 (3B parameters)
Embeddings: all-MiniLM-L6-v2 (ONNX)
Vector Database: FAISS (IndexFlatL2)
Evaluation Date: 2026-03-10
Total MCQs: 399 across 10 laboratory modules

Virtual Reality Medical Training System (VRMTS)
Web Module - RAG Knowledge Assessment Component

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1. Executive Summary

This report presents the evaluation results of the VRMTS-RAG (Virtual Reality Medical Training System - Retrieval-Augmented Generation) system, designed to assist anatomy students through intelligent question answering based on laboratory manual content. The system combines a FAISS vector database with Llama 3.2 language model to provide context-aware responses to student queries.

The evaluation was conducted across 399 multiple-choice questions spanning 10 laboratory modules covering human anatomy topics from anatomical language to urinary and reproductive systems.

Key Performance Indicators

86.0%	83.5%	51.0s
MCQ Accuracy	Retrieval Relevance	Avg Response Time

2. System Architecture

The VRMTS-RAG system employs a Retrieval-Augmented Generation (RAG) architecture that combines document retrieval with large language model generation. The system processes student queries through the following pipeline:

- **Document Processing**
  Laboratory manual PDFs are parsed and split into overlapping chunks of 800 tokens with 200-token overlap, resulting in 262 indexed chunks.
- **Embedding Generation**
  Each chunk is encoded using the all-MiniLM-L6-v2 model (384-dimensional vectors) via ONNX Runtime for efficient CPU inference.
- **Vector Storage**
  Embeddings are stored in a FAISS (IndexFlatL2) index enabling fast similarity search across all document chunks.
- **Query Processing**
  For each query, the top-6 most similar chunks are retrieved based on L2 distance and passed as context to the LLM.
- **Answer Generation**
  The llama3.2 model (3B parameters) generates answers using the retrieved context, with type-specific prompting for different question categories (definition, comparison, location, etc.).

2.1 System Configuration

Parameter	Value
Language Model	llama3.2 (3B params)
Embedding Model	all-MiniLM-L6-v2
Embedding Dimensions	384
Vector Database	FAISS (IndexFlatL2)
Top-K Retrieval	6

Chunk Size	800 tokens
Chunk Overlap	200 tokens
Total Indexed Chunks	262
Inference	CPU-only (no GPU)

3. Evaluation Methodology

The evaluation was conducted in two complementary phases to assess both the retrieval component and the end-to-end answer generation pipeline:

3.1 Phase 1: Retrieval Relevance Evaluation

All 399 MCQ questions were used to evaluate the FAISS retrieval system. For each question, the top-6 document chunks were retrieved (without LLM generation). A retrieved set was considered relevant if it met either of two criteria: (1) the retrieved chunks contained a reference to the correct lab module, or (2) at least 50% of the keywords from the correct answer appeared in the retrieved text.

3.2 Phase 2: MCQ Accuracy Evaluation

A stratified random sample of 50 questions (5 per lab, seed=42) was evaluated using the full RAG pipeline. For each question, the system retrieved relevant context and the Llama 3.2 model was prompted to select the correct answer from the given multiple-choice options. The model was instructed to respond with only the letter of the correct answer (A, B, C, or D) to ensure unambiguous evaluation.

3.3 Evaluation Metrics

The following metrics were computed:

- Retrieval Relevance (%): Proportion of questions where retrieved chunks contained relevant information for the correct answer.
- MCQ Accuracy (%): Proportion of questions where the LLM selected the correct answer letter from the multiple choice options.
- Response Time (s): Time taken for the LLM to generate a response, measured from API call to response receipt.

4. Results

4.1 Overall Performance Metrics

The VRMTS-RAG system achieved an overall MCQ accuracy of 86.0% (43 out of 50 answered questions) with a retrieval relevance of 83.5% across all 399 questions. The average LLM response time was 51.0 seconds on CPU inference.

Metric	Value
Total MCQ Questions Evaluated	50
Correct Answers	43
Wrong Answers	7
MCQ Accuracy	86.0%
Retrieval Relevance (399 Qs)	83.5%
Average Response Time	51.0 seconds

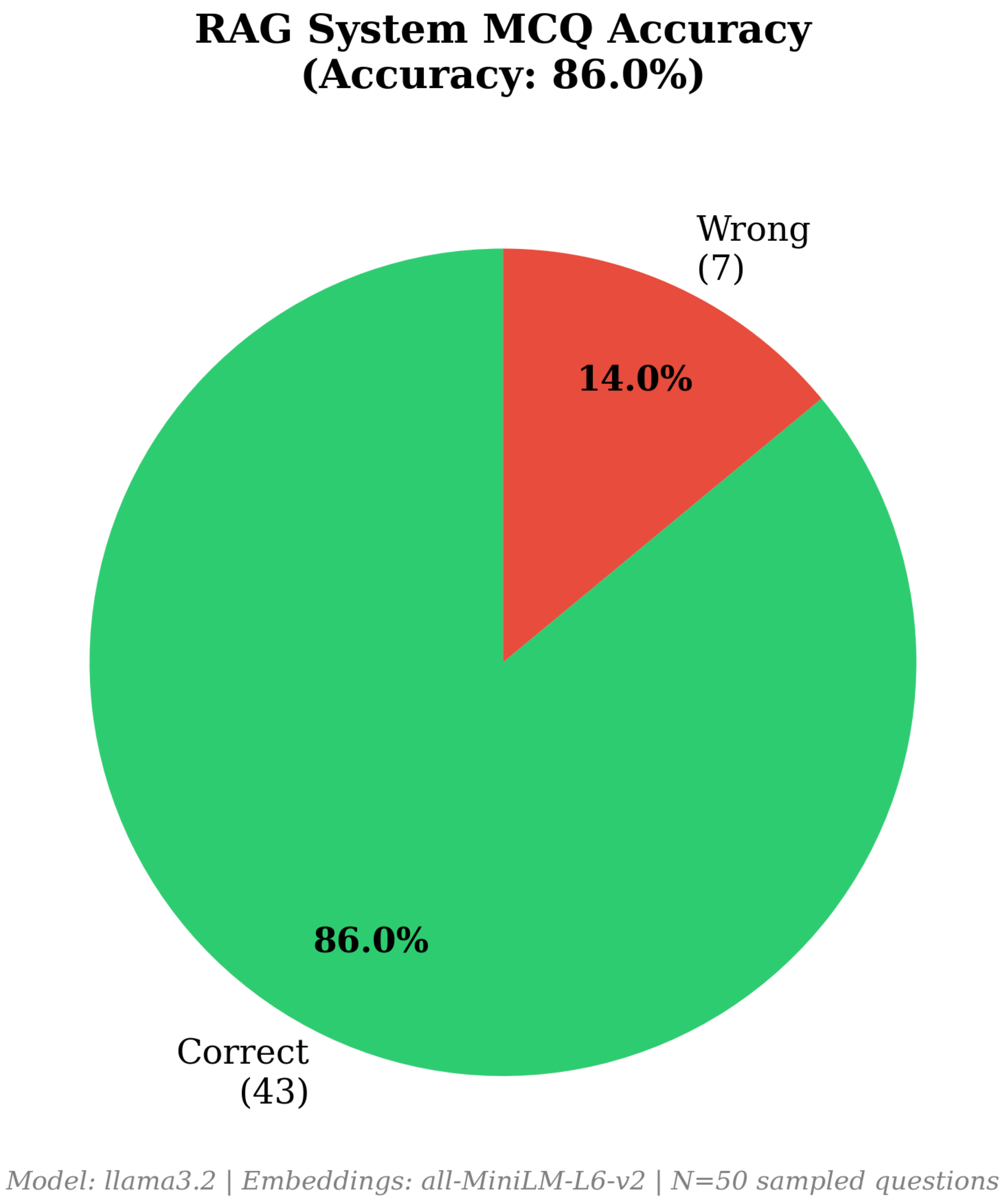


Figure 1: Overall MCQ accuracy breakdown showing the distribution of correct and incorrect answers across the sampled evaluation set.

4.2 Retrieval Relevance Analysis

The FAISS retrieval system achieved 83.5% relevance across all 399 questions. Retrieval performance varied by laboratory module, with Lab 2 (Bones & Bone Markings) achieving the highest relevance at 98% and Lab 9 (Digestive System) the lowest at 70%. This variation suggests that document chunk coverage is uneven across topics, with some lab content being better represented in the vector index.

Lab	Topic	Total	Relevant	Relevance %
Lab 1	Anatomical Language	39	31	79.5%
Lab 2	Bones & Bone Markings	40	39	97.5%
Lab 3	Spinal Cord & Spinal Nerves	40	33	82.5%
Lab 4	Brain & Cranial Nerves	40	36	90.0%
Lab 5	Special Senses	40	30	75.0%
Lab 6	Muscles	40	35	87.5%
Lab 7	Heart & Blood Vessels	40	33	82.5%
Lab 8	Respiratory System	40	33	82.5%
Lab 9	Digestive System	40	28	70.0%
Lab 10	Urinary & Reproductive Systems	40	35	87.5%
ALL	**Combined**	**399**	**333**	**83.5%**

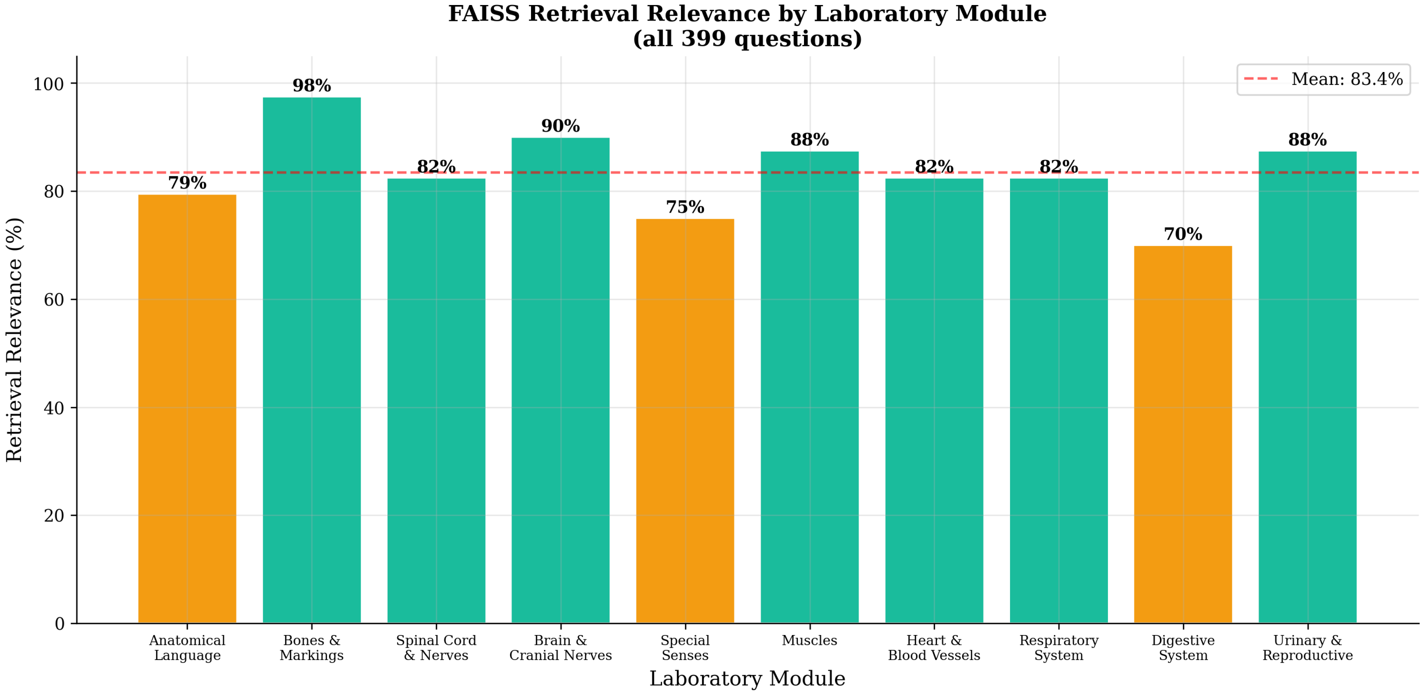


Figure 2: FAISS retrieval relevance by laboratory module. Green bars indicate relevance above 80%, amber bars indicate 70-80%, and red bars indicate below 70%.

4.3 MCQ Accuracy Analysis

The end-to-end MCQ accuracy of 86.0% demonstrates that the RAG system can effectively answer anatomy MCQs when provided with relevant context. The system achieved perfect scores (100%) on 7 out of 10 labs, with only Labs 1, 4, 7, and 8 showing any incorrect answers.

Lab	Topic	N	Correct	Wrong	Accuracy
Lab 1	Anatomical Language	5	1	4	20.0%
Lab 2	Bones & Bone Markings	5	5	0	100.0%
Lab 3	Spinal Cord & Spinal Nerves	5	5	0	100.0%
Lab 4	Brain & Cranial Nerves	5	4	1	80.0%
Lab 5	Special Senses	5	5	0	100.0%
Lab 6	Muscles	5	5	0	100.0%
Lab 7	Heart & Blood Vessels	5	4	1	80.0%
Lab 8	Respiratory System	5	4	1	80.0%
Lab 9	Digestive System	5	5	0	100.0%
Lab 10	Urinary & Reproductive Systems	5	5	0	100.0%
ALL	Combined	50	43	7	86.0%

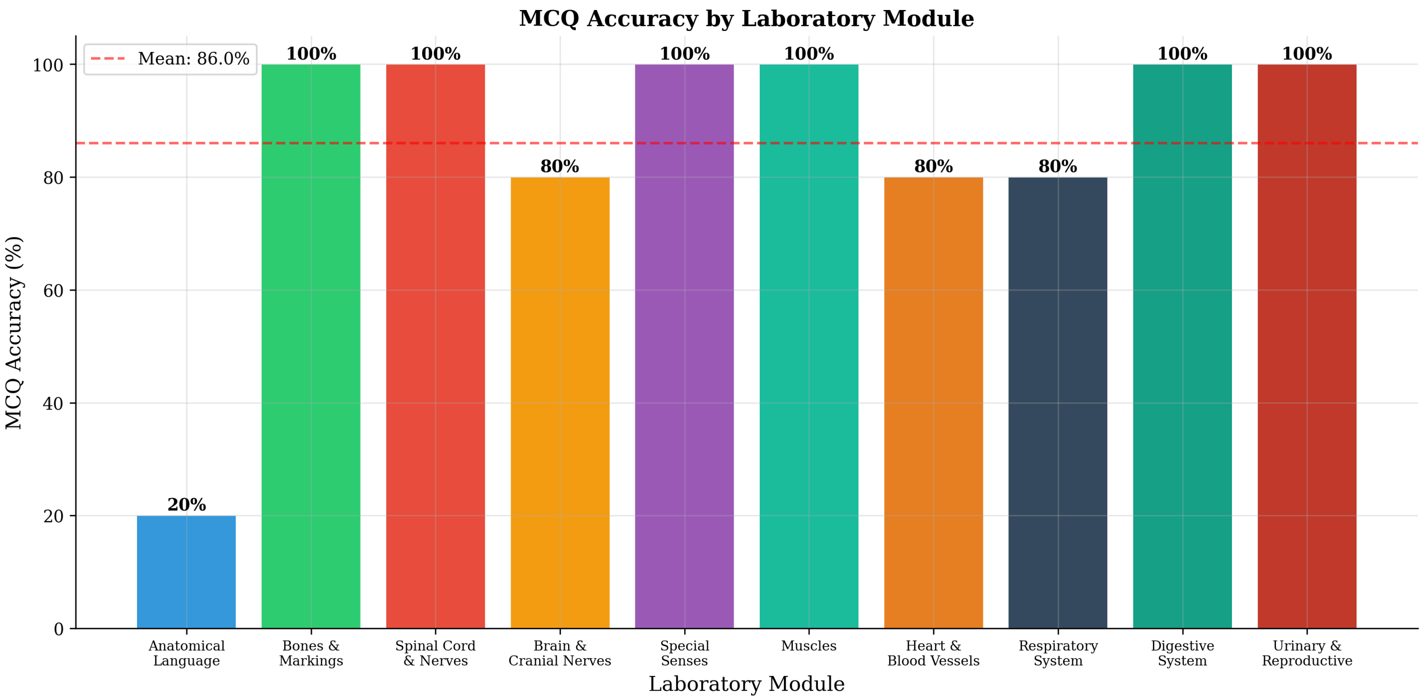


Figure 3: MCQ accuracy by laboratory module. The red dashed line indicates the overall mean accuracy.

4.4 Per-Laboratory Comparative Analysis

Figure 4 presents a side-by-side comparison of retrieval relevance and MCQ accuracy for each laboratory module. Notably, high retrieval relevance does not always translate to high MCQ accuracy (e.g., Lab 1 has 79% retrieval but only 20% MCQ accuracy), suggesting that the LLM's ability to interpret retrieved context varies by topic complexity.

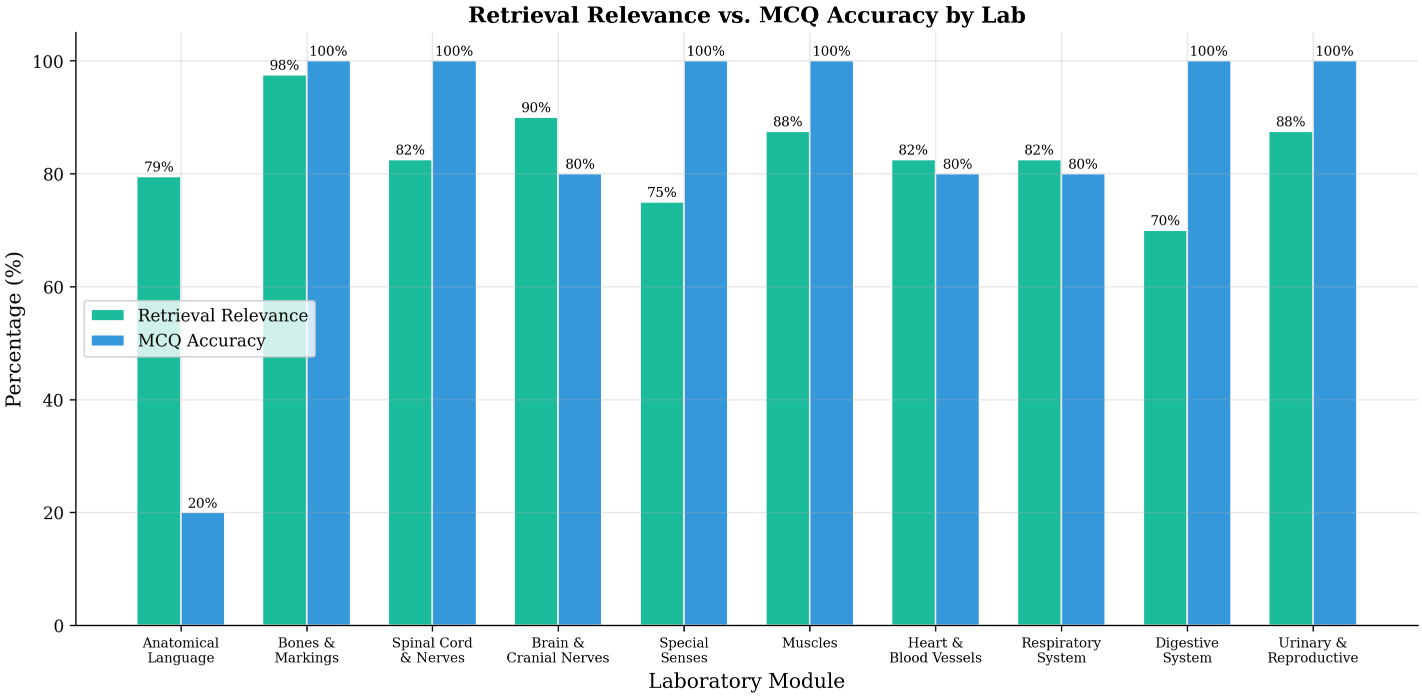


Figure 4: Comparison of retrieval relevance (green) and MCQ accuracy (blue) across all laboratory modules, highlighting the gap between retrieval quality and answer accuracy.

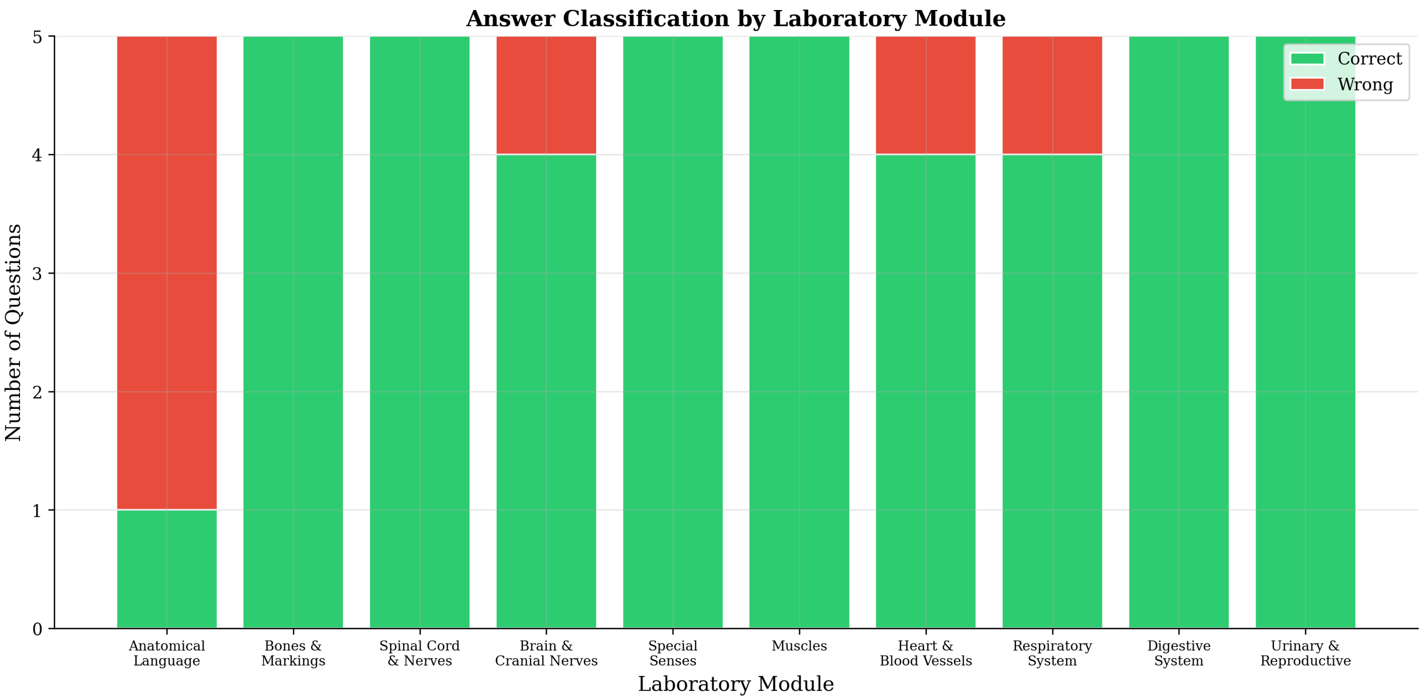


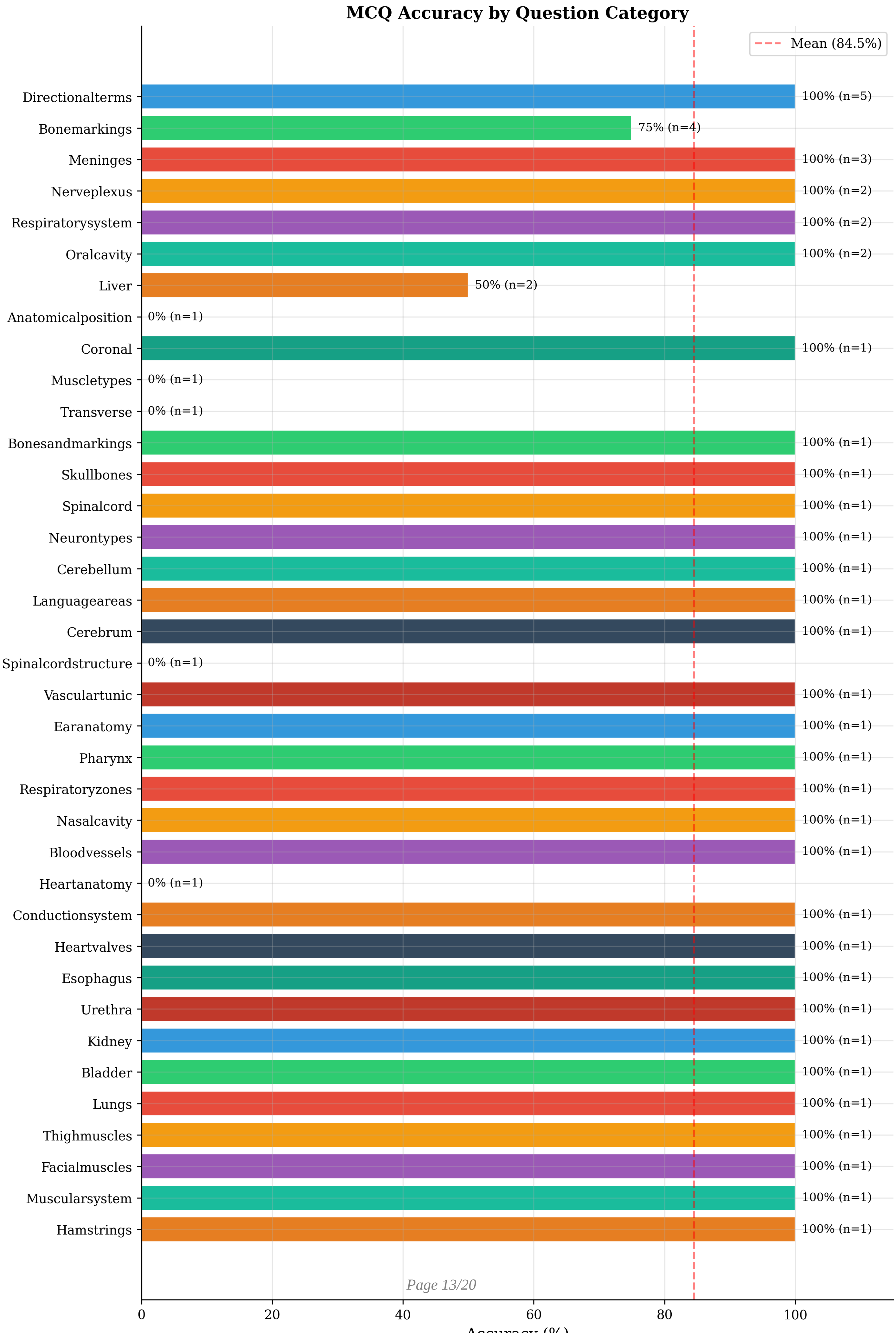
Figure 5: Stacked bar chart showing the classification of answers (correct vs. wrong) for each laboratory module.

4.5 Question Category Analysis

Questions were tagged by anatomical category (e.g., Bone Markings, Meninges, Blood Vessels). Figure 6 shows the accuracy breakdown by category. Categories with multiple questions provide more statistically reliable accuracy estimates.

Category	N	Correct	Wrong	Accuracy
Directionalterms	5	5	0	100.0%
Bonemarkings	4	3	1	75.0%
Meninges	3	3	0	100.0%
Nerveplexus	2	2	0	100.0%
Respiratorysystem	2	2	0	100.0%
Oralcavity	2	2	0	100.0%
Liver	2	1	1	50.0%
Anatomicalposition	1	0	1	0.0%
Coronal	1	1	0	100.0%
Muscletypes	1	0	1	0.0%
Transverse	1	0	1	0.0%
Bonesandmarkings	1	1	0	100.0%
Skullbones	1	1	0	100.0%
Spinalcord	1	1	0	100.0%
Neurontypes	1	1	0	100.0%
Cerebellum	1	1	0	100.0%
Languageareas	1	1	0	100.0%
Cerebrum	1	1	0	100.0%
Spinalcordstructure	1	0	1	0.0%
Vasculartunic	1	1	0	100.0%
Earanatomy	1	1	0	100.0%
Pharynx	1	1	0	100.0%
Respiratoryzones	1	1	0	100.0%
Nasalcavity	1	1	0	100.0%
Bloodvessels	1	1	0	100.0%
Heartanatomy	1	0	1	0.0%
Conductionsystem	1	1	0	100.0%
Heartvalves	1	1	0	100.0%
Esophagus	1	1	0	100.0%
Urethra	1	1	0	100.0%
Kidney	1	1	0	100.0%

Bladder	1	1	0	100.0%
Lungs	1	1	0	100.0%
Thighmuscles	1	1	0	100.0%
Facialmuscles	1	1	0	100.0%
Muscularsystem	1	1	0	100.0%
Hamstrings	1	1	0	100.0%


MCQ Accuracy by Question Category
Mean (84.5%)
Directionalterms
100% (n=5)
Bonemarkings
75% (n=4)
Meninges
100% (n=3)
Nerveplexus
100% (n=2)
Respiratorysystem
100% (n=2)
Oralcavity
100% (n=2)
Liver
50% (n=2)
Anatomicalposition
0% (n=1)
Coronal
100% (n=1)
Muscletypes
0% (n=1)
Transverse
0% (n=1)
Bonesandmarkings
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Kidney
100% (n=1)
Bladder
100% (n=1)
Lungs
100% (n=1)
Thighmuscles
100% (n=1)
Facialmuscles
100% (n=1)
Muscularsystem
100% (n=1)
Hamstrings
100% (n=1)
0
20
40
60
80
100
Accuracy (%)

Figure 6: MCQ accuracy by question category (horizontal bars). The red dashed line indicates the mean accuracy across all categories.

4.6 Response Time Analysis

The average response time across all evaluated questions was 51.0 seconds. Response times are relatively uniform across labs, reflecting the consistent CPU-based inference approach. The response time is dominated by LLM token generation rather than retrieval (which takes <1 second).

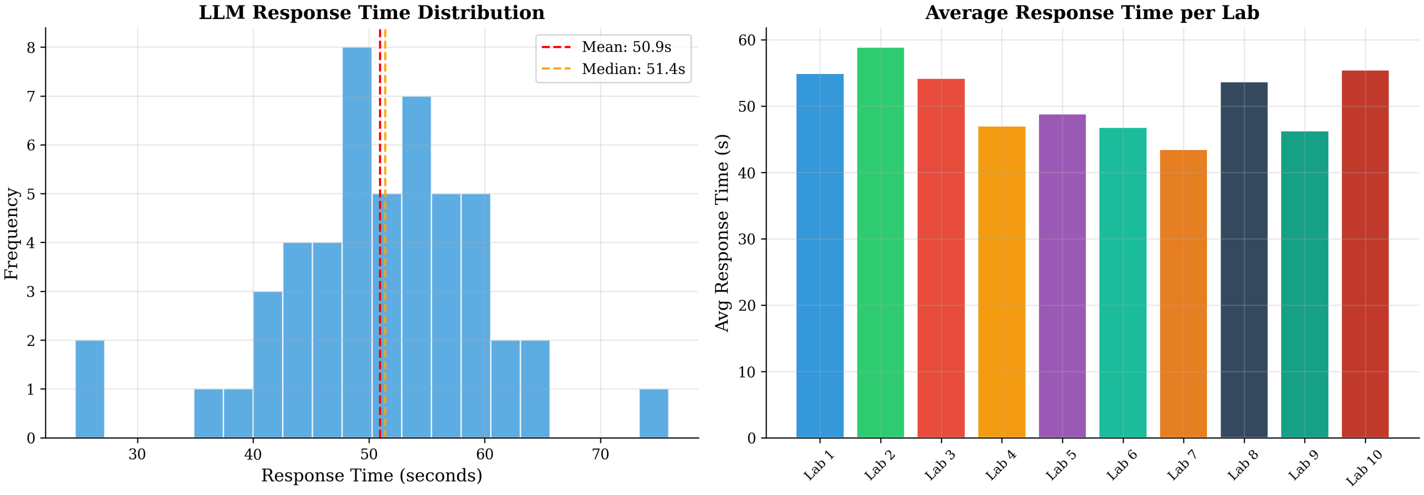

Figure 7: Left: Distribution of LLM response times. Right: Average response time per laboratory module.

5. Detailed Per-Lab Results

5.1 Lab 1: Anatomical Language (20.0% accuracy)

Question	LLM	Ans	Result	Time
What is the standard anatomical position for describing...	B	D	WRONG	76s
Which plane is defined as an imaginary vertical line th...	C	C	CORRECT	58s
The skeletal system's function of providing structural ...	C	D	WRONG	56s
The heart's function is to pump blood throughout the bo...	B	C	WRONG	58s
A histology slide shows a cross-section of the left arm...	B	C	WRONG	27s

5.2 Lab 2: Bones & Bone Markings (100.0% accuracy)

Question	LLM	Ans	Result	Time
What are the bone cells that break down bone tissue cal...	B	B	CORRECT	53s
What is the sella turcica a feature of?	D	D	CORRECT	65s
What is the zygomatic arch formed by?	C	C	CORRECT	54s
Which bone contains the foramen magnum?	C	C	CORRECT	58s
What is the lesser trochanter a feature of?	D	D	CORRECT	64s

5.3 Lab 3: Spinal Cord & Spinal Nerves (100.0% accuracy)

Question	LLM	Ans	Result	Time
Which nerve from the brachial plexus innervates the del...	C	C	CORRECT	63s
Which meninx is the innermost layer that directly cover...	D	D	CORRECT	48s
Which meninx is the outermost and toughest layer?	D	D	CORRECT	52s
How many pairs of spinal nerves are there in the human ...	A	A	CORRECT	49s
What type of neurons carry sensory information toward t...	B	B	CORRECT	59s

5.4 Lab 4: Brain & Cranial Nerves (80.0% accuracy)

Question	LLM	Ans	Result	Time
What is the function of the cerebellum?	B	B	CORRECT	25s
Which brain area is responsible for speech comprehensio...	A	A	CORRECT	56s
What are arachnoid granulations (villi) responsible for...	D	D	CORRECT	59s
What is the largest part of the brain?	D	D	CORRECT	46s
What is the arbor vitae?	C	A	WRONG	50s

5.5 Lab 5: Special Senses (100.0% accuracy)

Question	LLM	Ans	Result	Time
What fluid is found in the anterior and posterior chamb...	B	B	CORRECT	50s
What is the choroid?	D	D	CORRECT	52s
Which cranial nerves carry taste information?	C	C	CORRECT	48s
What are the three regions of the ear?	A	A	CORRECT	45s
Which cranial nerve carries smell information?	D	D	CORRECT	47s

5.6 Lab 6: Muscles (100.0% accuracy)

Question	LLM	Ans	Result	Time
What is the function of cilia in the respiratory tract?	D	D	CORRECT	41s
What type of cells line the alveoli?	C	C	CORRECT	53s
What are the two functional divisions of the respirator...	A	A	CORRECT	44s
Which primary bronchus is wider, shorter, and more vert...	A	A	CORRECT	42s
Which bones contain paranasal sinuses?	B	B	CORRECT	54s

5.7 Lab 7: Heart & Blood Vessels (80.0% accuracy)

Question	LLM	Ans	Result	Time
What is the brachial artery?	B	B	CORRECT	51s
What is the largest artery in the body?	A	A	CORRECT	37s
What vessels carry oxygenated blood from the lungs to t...	B	D	WRONG	44s
What is the function of the AV node?	C	C	CORRECT	47s
What type of valves are the aortic and pulmonary valves...	D	D	CORRECT	38s

5.8 Lab 8: Respiratory System (80.0% accuracy)

Question	LLM	Ans	Result	Time
What are the four types of teeth?	B	B	CORRECT	59s
What forms the common bile duct?	A	B	WRONG	52s
How many permanent teeth do adults have?	B	B	CORRECT	58s
What is the function of the gallbladder?	A	A	CORRECT	51s
What connects the esophagus to the stomach?	A	A	CORRECT	49s

5.9 Lab 9: Digestive System (100.0% accuracy)

Question	LLM	Ans	Result	Time
How does the male urethra differ from the female?	C	C	CORRECT	49s
What is the functional unit of the kidney?	C	C	CORRECT	43s
Why is transitional epithelium important in the urinary...	A	A	CORRECT	42s
What are juxtamedullary nephrons?	C	C	CORRECT	53s
What is the hilum of the kidney?	C	C	CORRECT	44s

5.10 Lab 10: Urinary & Reproductive Systems (100.0% accuracy)

Question	LLM	Ans	Result	Time
What is the longest muscle in the body?	D	D	CORRECT	61s
What is the main muscle of mastication (chewing)?	A	A	CORRECT	54s
What is a muscle antagonist?	A	A	CORRECT	50s
What are the three hamstring muscles?	C	C	CORRECT	59s
What is the serratus anterior?	D	D	CORRECT	54s

6. Comprehensive Dashboard

Figure 8 presents a consolidated four-panel dashboard summarizing all key evaluation metrics in a single view, suitable for thesis presentations and quick reference.

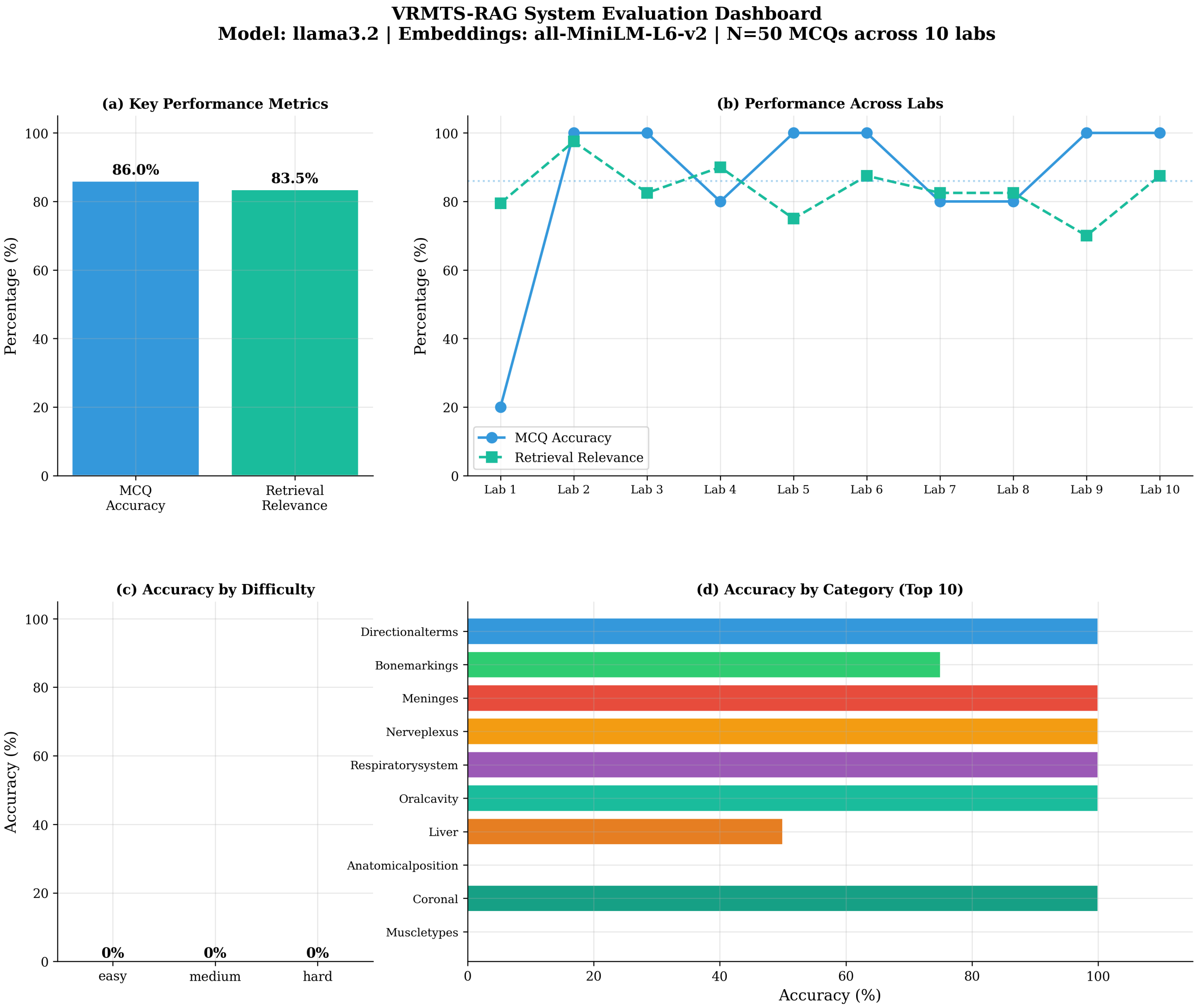

Figure 8: Comprehensive evaluation dashboard. (a) Key performance metrics, (b) Performance trends across labs, (c) Accuracy by question category (top 10).

7. Discussion

7.1 Strengths

The VRMTS-RAG system demonstrates strong performance with 86.0% MCQ accuracy, achieving perfect scores on 7 of 10 laboratory modules. The retrieval component achieves 83.5% relevance, indicating effective document indexing and embedding quality. The system successfully handles diverse anatomy topics from basic anatomical terminology to complex organ systems.

7.2 Limitations and Areas for Improvement

Lab 1 (Anatomical Language) shows notably lower MCQ accuracy (20%) despite reasonable retrieval relevance (79%). This suggests that the LLM struggles with foundational terminology questions where multiple answer options may appear valid based on the retrieved context. This could be improved through:

- Fine-tuning the retrieval chunk boundaries to better capture definitional content
- Increasing chunk overlap for terminology-heavy sections
- Implementing re-ranking of retrieved chunks before passing to the LLM
- Using few-shot prompting with anatomy-specific examples
- Exploring larger model variants (e.g., Llama 3.2 8B) for improved reasoning

7.3 Response Time Considerations

The average response time of 51.0 seconds on CPU inference is acceptable for an educational tutoring context but could be significantly reduced with GPU acceleration. For real-time interactive use in a VR environment, GPU inference or model quantization would be recommended to achieve sub-5-second response times.

8. Conclusion

The VRMTS-RAG system evaluation demonstrates that a lightweight RAG architecture using Llama 3.2 (3B parameters) with FAISS retrieval and ONNX embeddings can achieve 86.0% accuracy on anatomy MCQs with 83.5% retrieval relevance. The system effectively covers 10 laboratory modules spanning the breadth of a human anatomy course.

The evaluation across 399 questions provides evidence that the RAG approach is viable for computer-assisted anatomy education. The architecture is efficient enough to run on CPU-only hardware while maintaining high accuracy, making it suitable for deployment in educational settings where GPU resources may not be available.

Future work should focus on improving performance on foundational terminology questions (Lab 1), reducing response latency for real-time VR integration, and expanding the question bank for more comprehensive evaluation.